
EXERCISES

Exercise sheet 7

AI504 — Knowledge Representation

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1 Exercise 1*

Consider the Boolean objective function:

$$f(x) = x_1 \wedge (x_2 \vee \neg x_3) \wedge (\neg x_1 \vee \neg x_2)$$

Formulate the problem as an integer linear program. Can any Boolean satisfiability problem be formulated as an integer linear program?

1.1 Solution

$$\begin{array}{ll} \max_x & f(x) \\ \text{s.t.} & Ax = b \\ & x \in \mathbb{Z}^n \end{array}$$

2 Exercise 2*

Consider $\max\{c^\top x \mid Ax = b, x \in \mathbb{Z}^n\}$. Sometimes the solution of the linear relaxation is already integral. Can you find a sufficient condition for the matrix A for that to happen?

2.1

$$\begin{array}{ll} \max_x & c^\top x \\ \text{s.t.} & Ax = b \\ & x \in \mathbb{R}^n \end{array}$$

Firstly A must only have integer entries

Also for

$$A_B x_B + A_N x_N = b \Rightarrow x = A_B^{-1} b$$

Since for $x \in \mathbb{Z}$ then

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

For this A_B must have $\det(A) = 0, -1, 1$

“ A should be a totally unimodular matrix”

3 Exercise 3*

Consider the following problem:

$$\begin{aligned} \max_x \quad & x_1 - x_2 \\ \text{s.t.} \quad & \frac{2}{3}x_1 + \frac{1}{2}x_2 \leq 1 \\ & x_1, x_2 \geq 0 \\ & x_1, x_2 \in \mathbb{Z} \end{aligned}$$

Derive a Chvatal-Gomory cut.

Since $x_{\text{LP}}^* = (\frac{3}{2}, 0)$

The cut should be $x_1 \leq 1$

4 Exercise 4*

The 0-1 knapsack problem is a combinatorial optimization problem that can be described as follows: Given a set of items, each with a size/weight and a value, the problem is to choose the items that maximize the total value under the condition that the total size/weight is below a certain threshold.

Design and apply by hand a dedicated branch and bound algorithm to the following instance of the 0-1 knapsack problem: values $v = [9, 4, 2, 3, 5, 3]$, weights $w = [7, 8, 4, 5, 9, 4]$ and capacity $W = 20$.

You will find useful the following observation that can be proven true: The relaxed knapsack problem can be efficiently solved with a greedy approach. Items are added one at a time by selecting the next item with the greatest ratio of value to weight. If there is enough remaining capacity, the item is fully assigned with $x_i = 1$. If not, a fractional value is assigned such that the remaining capacity is saturated and all remaining items have $x_i = 0$.

item:	1	2	3	4	5	6
value:	9	4	2	3	5	3
weight:	7	8	4	5	9	4
ratio:	9/7	4/8	2/4	3/5	5/9	3/4
ratio:	1.28	0.5	0.5	0.6	0.555	0.75

Figure 1: table form exercise

4.1 Solution

$$\begin{aligned} & \max \sum_i v_i x_i \\ \text{s.t. } & \sum_i w_i x_i \leq W \\ & x_i \in \end{aligned}$$

This means that $\text{obj} = 17$

sort by $\frac{v}{w}$

Then we do greedy search to fit for 17

5 Exercise 6*

A medley relay is a team swimming event where each swimmer on the team swims a different stroke in a specific order: backstroke, breaststroke, butterfly, and freestyle. This event can vary in distance, typically ranging from 100 to 400 meters, depending on the competition. Consider the problem of selecting students for a swimming medley relay team. In Figure 2 show times for each swimming style of five students.

Student	backstroke	breaststroke	butterfly	freestyle
A	43.5	47.1	48.4	38.2
B	45.5	42.1	49.6	36.8
C	43.4	39.1	42.1	43.2
D	46.5	44.1	44.5	41.2
E	46.3	47.8	50.4	37.2

Figure 2: table from exercise

We need to choose a student for each of the four swimming styles such that the total relay time is minimized. Try first to do this task by hand. Then, formulate the problem as a MILP and solve it in Python. Finally, compare the solution with the one obtained by hand.

5.1 Solution

backstroke: A breaststroke: C butterfly: D freestyle: B

$$43.5 + 39.1 + 44.5 + 36.8 = 163.9$$

Let $j \in \{1, 2, 3, 4\}$ and $i \in \{A, B, C, D, E\}$

then $A = []$

$$\begin{aligned}
& \min \sum_{ij} t_{ij} x_{ij} \\
s.t \quad & \sum_{i=1}^5 x_{ij} = 1, \quad \forall j \\
& \sum_{j=i}^4 x_{ij} \leq 1 \quad \forall i \\
& x_{ij} \in \{0, 1\}
\end{aligned}$$